

A Sharp Black-Box Barrier from Factorwise Fourier Bounds to a Product Twist

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Abstract

Rudin–Shapiro coefficients give two factors with uniform square-root Fourier bounds whose modulated pointwise product has a full linear resonance at a prescribed atom. This stops only the factorwise-single-Fourier black-box implication. The classification is `METHOD_OBSTRUCTION_L1` and the verdict is `STOP_SCOPED`. No program-positive L2 claim or endpoint exponent credit is made.

1 Target contract and source boundary

The target has six simultaneous axes: actual fixed- h_0 packet, source-locked named physical atom, every deterministic prefix, every deterministic scale, a fixed- X power at that atom, and actual active support. The value $h_0 = 2$ is source-backed data only. Repository hashes are integrity locks, not theorem evidence. TPC-193’s declared seven-source corpus remains `STOP-SCOPED` [1].

2 Rudin–Shapiro witness

Let $P_0 = Q_0 = 1$ and recursively set

$$P_{m+1}(z) = P_m(z) + z^{2^m} Q_m(z), \quad Q_{m+1}(z) = P_m(z) - z^{2^m} Q_m(z).$$

The coefficients of P_m are signs ρ_n , $0 \leq n < L = 2^m$, and for $|z| = 1$,

$$|P_m(z)|^2 + |Q_m(z)|^2 = 2L.$$

Consequently every additive Fourier sum of $f(n) = \rho_n$ is at most $\sqrt{2L}$.

Theorem 1 (Sharp black-box failure). *Fix any $\alpha_\star$. Put $g(n) = \rho_n e(\alpha_\star n)$. Both f and g have uniform $O(\sqrt{L})$ Fourier bounds, but*

$$\sum_{n < L} f(n)g(n)e(-\alpha_\star n) = L.$$

Therefore factorwise uniform Fourier bounds, used as black-box premises, do not imply cancellation of the product at a prescribed atom.

Proof. Modulation translates the Fourier variable, so g inherits the same supremum bound. Since $\rho_n^2 = 1$, every term in the displayed product sum equals one. $\square$

3 Scoped interpretation

The witness is synthetic and is not asserted to model the Möbius function. It rejects exactly the logical implication

two independent one-factor Fourier bounds $\implies$ literal two-factor product bound.

Arithmetic identities that couple the two Möbius factors remain open.

4 Loss, level and scope ledger

The smallest missing literal theorem is

ARITHMETIC_TWO_FACTOR_COUPLING_THEOREM.

The next route is TWO_MOBIUS_PRODUCT_FOURIER_ROUTE. The cell FACTORWISE_SINGLE_MOBIUS_FOURIER_TO_LITERAL_PRODUCT is STOP-SCOPED; neither O161 parent nor the global architecture is stopped.

The named-atom exponent credit is 0, while the endpoint requires a strict budget greater than $1/400$; the ledger is unpaid. Block sums are not cumulative prefixes, symbolic packet atoms are not named production atoms, phase L^2 and almost-everywhere statements are not pointwise evaluation, and scoped method failures are not theorem or architecture failures.

5 Machine certificate

The adjacent canonical payload freezes the formula or finite witness, exact repository-source hashes, any explicitly non-hashed external locator record, six target axes, scoped stop, route state and claim firewall. Its recursive exact schema has closed objects, exact array positions and constant leaves. The checker recomputes the finite certificate and executes ten adversarial mutations.

References

- [1] Liang Wang. Literal fixed-atom candidate mechanism gate, 2026.